



Problem solving with fractions and decimals

Task A: Solving problems with fractions

1) Arrange five different digits into these inequalities.

a) *E.g.*

$$\boxed{5} > \frac{\boxed{7}}{\boxed{2}} > \frac{\boxed{9}}{\boxed{3}}$$

b) *E.g.*

$$\frac{\boxed{5}}{\boxed{2}} > \frac{\boxed{4}}{\boxed{3}} > \boxed{1}$$

2) Arrange the digits 1 to 9 to make these inequalities true.

a) *E.g.*

$$\boxed{1} > \frac{\boxed{8}}{\boxed{9}} > \frac{\boxed{6}}{\boxed{7}} > \frac{\boxed{4}}{\boxed{5}} > \frac{\boxed{2}}{\boxed{3}}$$

b) *E.g.*

$$\frac{\boxed{3}}{\boxed{2}} > \frac{\boxed{5}}{\boxed{4}} > \frac{\boxed{7}}{\boxed{6}} > \frac{\boxed{9}}{\boxed{8}} > \boxed{1}$$

3) Izzy, Andeep, Lucas, Alex and Sam record their scores for a spelling competition.

Rank them from worst to best.

Izzy	$\frac{11}{12}$
Andeep	$\frac{4}{5}$
Lucas	$\frac{5}{6}$
Alex	$\frac{18}{20}$
Sam	$\frac{26}{30}$

*Ranked:
Andeep, Lucas, Sam, Alex, Izzy*



Task B: Solving problems with decimals

1) Here are the prices of some items in a cafe.



a) Write them in order from cheapest to most expensive.



b) You have a £5.00 gift voucher but you won't get any change.

You therefore want to spend as close to £5.00 as possible.

What can you buy?

A sandwich and a milkshake. Total cost = £2.95 + £1.98 = £4.93

c) What is the largest number of items you can buy with £5.00 and what are they?

An apple, a banana, a juice and a cupcake.

Total cost = £0.77 + £0.76 + £1.54 + £1.58 = £4.65



Task C: Solving problems with fractions and decimals

- 1) Work out the length of ribbon needed for cushions with these dimensions. (All dimensions are in metres)

Width	Length	Converted length / width	Calculation	Length needed
0.64	$\frac{9}{20}$	0.45	E.g. $2(0.64 + 0.45)$	2.18 m
1.38	$\frac{3}{5}$	0.6	E.g. $2(1.38 + 0.6)$	3.96 m
0.15	$\frac{3}{25}$	0.12	E.g. $2(0.15 + 0.12)$	0.54 m
$\frac{7}{20}$	1.07	0.35	E.g. $2(0.35 + 1.07)$	2.84 m
$\frac{7}{8}$	0.97	0.875	E.g. $2(0.875 + 0.97)$	3.69 m

- 2) A stopwatch measures time in minutes and seconds:

20:50

- a) Is this more or less than 20.5 minutes?
By how much?

It is 20 seconds more than 20.5 minutes

20:30

- b) Is this more or less than 20.3 minutes?
By how much?

It is 12 seconds more than 20.3 minutes

- c) Which time is closer to $20\frac{5}{10}$ minutes? *20:30 is exactly equal to $20\frac{5}{10}$ minutes.*

- d) Which time is closer to $20\frac{1}{3}$ minutes? *20:30 is closer to $20\frac{1}{3}$ minutes (as this would be 20:20).*

- e) Which do you think are further apart: 20.4 minutes and $20\frac{1}{4}$ minutes, or 20.5 minutes and $20\frac{1}{5}$ minutes.

20.5 minutes and $20\frac{1}{5}$ minutes are furthest apart. For 20.4 minutes, 20:24 would be showing and, for $20\frac{1}{4}$ minutes, 20:15 would be showing. These are nine seconds apart. For 20.5 minutes, 20:30 would be showing and, for $20\frac{1}{5}$ minutes, 20:12 would be showing. These are 18 seconds apart.

